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TEXT CLASSIFICATION IN NLP: MEDIA BIAS DETECTION, [STUDENT 2 TOPIC], AND [STUDENT 3 TOPIC]

by

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Submitted for the course

CMPE346: NATURAL LANGUAGE PROCESSING

for the assignment

FINAL PROJECT

June 2026

Abstract

In this paper, we performed text classification on three different NLP tasks using an identical deep learning model. Each task is an application of classification in the NLP field: Media bias detection of news articles, [Student 2 task], and [Student 3 task]. Three models are applied to each task, including one traditional deep learning model Bidirectional LSTM, and two transformer-based pre-trained models BERT and RoBERTa. The three models are then fine-tuned on the dataset for each specific task, and accuracy and weighted F1-score are used as evaluation metrics. For all three tasks, we confirmed that transformer-based models outperform Bidirectional LSTM, and RoBERTa attains the best results.

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Introduction

Text classification is one of the foundational tasks in NLP and underpins many practical applications we use on a daily basis. At its most basic, text classification assigns a label from a predetermined set to an instance of text based on its contents, providing a pathway to automating understanding over large scales. These applications include, but are not limited to: finding misinformation and media bias in news; classifying the argumentative intent of scientific documents; and classifying harmful content in social media.

While the formulation of text classification may be seemingly simple (the assignment of a discrete label to an item of text), the complexities of human language provide a variety of challenges to be addressed in text classification such as learning the semantics and contextuality of words. Early techniques relied on manually engineered features, shallow machine learning models such as Naive Bayes [1] and Support Vector Machines [2], and more recently recurrent neural networks, particularly Long Short-Term Memory networks [3], began learning representations directly from the text. Most impactfully however, was the introduction of the Transformer [4] and subsequently, the pre-training and subsequent release of very large scale language models such as BERT [5] and RoBERTa [6].

This project looks at text classification in three different NLP tasks each corresponding to a different area of application. The group then proceeds to individually train three different models (BiLSTM, BERT, RoBERTa) on three individually chosen datasets and reports their performance in terms of accuracy and weighted F1-score. The intention of this is to compare how the three models behave under different circumstances.

Chapter 1

Atakan Gül — Media Bias Detection in News

1.1 Methodology

This work targets the task of media bias detection from news articles. It is a classification task where the model has to decide whether a news sentence has biased framing or no bias. Media bias detection is a non-trivial NLP problem because bias is manifested by subtler things like phrasing, framing and tone, and not by stating false facts [7].

In this work, 3 models are tested:

Bidirectional LSTM. A Long Short-Term Memory network is a type of recurrent neural network designed to remember information for long periods of time [3]. Bidirectional LSTMs are trained in both directions in order to make better use of the context. In our work the model takes an embedding layer, two layers of 256-hidden-dimension Bidirectional LSTMs and a linear classification layer on top of the concatenation of the last forward and backward hidden states.

BERT. BERT is a pre-trained language model based on the Transformer encoder architecture. It is trained on a large corpus using a masked language modeling objective and a next sentence prediction objective and therefore is able to produce deep bidirectional context representations [5]. We feed the [CLS] token output through a linear classifier. For the implementation of BERT we use the `bert-base-uncased` variant.

RoBERTa. RoBERTa is a BERT variant that removes next sentence prediction, uses dynamic masking, and is trained with larger corpora and batch sizes [6]. This architecture has been shown to perform consistently better on downstream tasks. For RoBERTa we use the `roberta-base` vari-

ant.

The transformer models (BERT and RoBERTa) are fine-tuned from the beginning with an AdamW optimizer and a linear learning rate scheduler with warmup. The LSTM model is trained from scratch using the Adam optimizer.

1.2 Experiments & Findings

Dataset. The BABE dataset (Bias Annotations By Experts), introduced by Spinde et al. [8], is employed for this task. The Hugging Face distribution used in this project contains 4,121 news sentences gathered from a diverse collection of sources, each rated as either biased (1) or not biased (0) by trained annotators; it is divided into 3,121 training and 1,000 test samples [9]. Since the labeling is conducted with the help of expert annotators and has been designed carefully, it serves as a reliable reference for media bias detection.

Training Configuration. For all models, 3 epochs of training are performed with early stopping on F1 score on the validation set, a stopping threshold of 0.5% difference between consecutive performances and a patience of 2 epochs. A batch size of 16 and a maximum sequence length of 128 are set. The learning rate for transformers is 2×10^{-5} and for the LSTM is 1×10^{-3} . All models are trained on an NVIDIA GeForce RTX 3060 Laptop GPU.

Results. Table 1.1 presents the performance of each model on the BABE test set.

Model	Accuracy	F1
LSTM	0.6560	0.6511
BERT	0.8260	0.8266
RoBERTa	0.8460	0.8464

Table 1.1: Model performance on the BABE dataset for media bias detection.

Figure 1.1 provides a visual comparison of accuracy and F1 scores across all three models. Figure 1.2 shows the training loss per epoch for each model. Figure 1.3 presents the confusion matrices on the test set.

Qualitative Examples. Table 1.2 presents selected test sentences illustrating correct predictions, the transformer advantage over LSTM, and disagreements between BERT and RoBERTa.

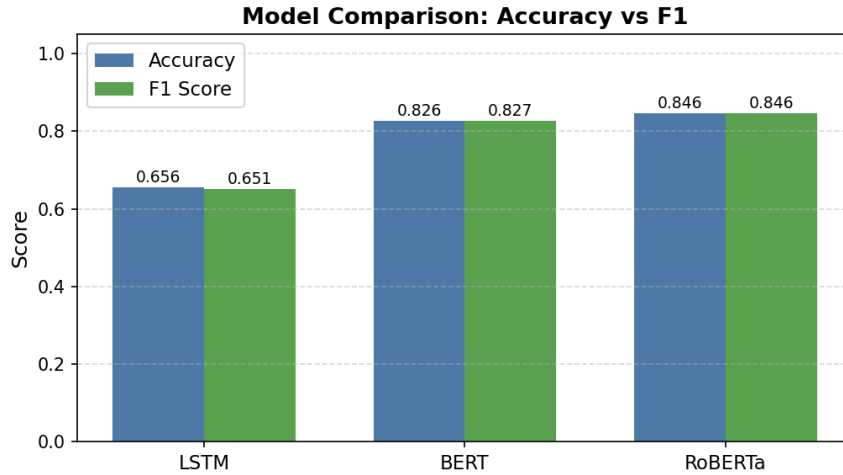


Figure 1.1: Accuracy and F1 score comparison across LSTM, BERT, and RoBERTa.

1.3 Discussion

From these figures, it is clear that there is a performance hierarchy across the three architectures. The Bidirectional LSTM achieves 65.6% accuracy and an F1 score of 0.651, which, while above random chance, falls well short of the transformer-based models. This gap reflects the difficulty of learning both language representations and task-specific patterns from scratch on a relatively small dataset.

BERT scores well, gaining 82.6% accuracy and an F1 score of 0.827, demonstrating the power of transfer learning. The confusion matrix in Figure 1.3 shows that BERT handles both classes much more reliably than the LSTM, with fewer misclassifications on biased sentences.

RoBERTa achieves 84.6% accuracy and 0.846 F1 score. The RoBERTa paper attributes its downstream improvements to a more carefully optimized pre-training procedure, including the removal of the next sentence prediction objective, dynamic masking, larger batches, and more training data [6]. The training loss curve in Figure 1.2 shows that RoBERTa converges more smoothly than BERT, which is consistent with its more robust pretraining. The results in this experiment are consistent with that expectation, since RoBERTa is the strongest of the three tested architectures.

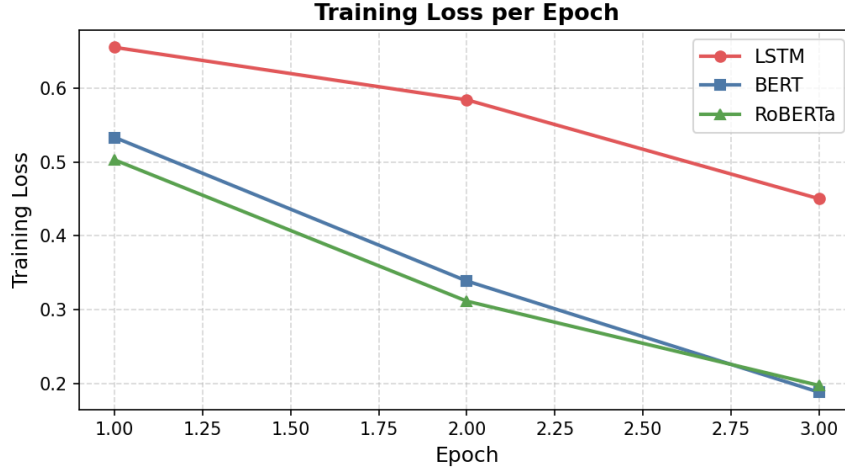


Figure 1.2: Training loss per epoch for each model.

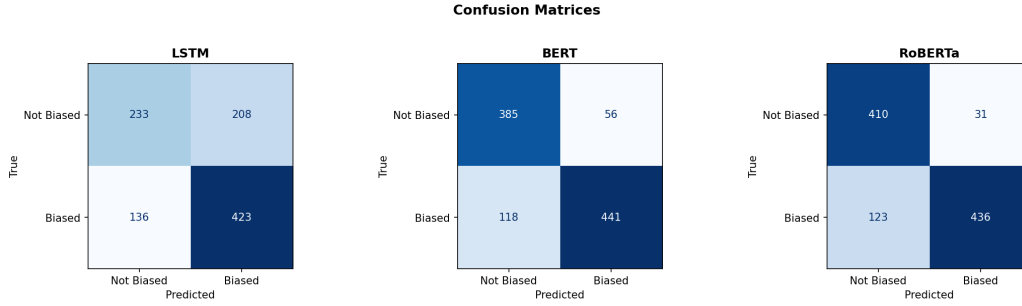


Figure 1.3: Confusion matrices on the BABE test set for each model.

1.4 Conclusion

In this paper, three different deep learning architectures were implemented for media bias detection on the BABE dataset. The experimental results showed that the pretrained transformers performed significantly better than the LSTM baseline, and the best model was RoBERTa with 84.6% accuracy and 0.846 weighted F1-score. The experiments also showed that pretraining plays an important role for low-resource classification tasks, where learning the language nuances of news framing and bias from scratch is difficult. Future work can use larger transformer models or domain-adaptive pretraining on news corpora.

Sentence	True	LSTM	BERT	RoBERTa
<i>“Nearly 78% of Americans report experiencing concern about the climate...”</i>	NB	NB ✓	NB ✓	NB ✓
<i>“Companies like Nike and Google have made Juneteenth a paid holiday.”</i>	NB	B ×	NB ✓	NB ✓
<i>“Rahaf Mohammed al-Qununi’s case drew global attention to Saudi Arabia’s strict social rules...”</i>	NB	B ×	NB ✓	NB ✓
<i>“None of Biden’s Democrat counterparts challenged his claim of economic selflessness.”</i>	B	B ✓	NB ×	B ✓

Table 1.2: Qualitative prediction examples. NB = Not Biased, B = Biased. ✓ = correct, × = incorrect.

Chapter 2

[Student 2 Name] — [Student 2
Topic]

2.1 Methodology

[Student 2 fills this in]

2.2 Experiments & Findings

[Student 2 fills this in]

2.3 Discussion

[Student 2 fills this in]

2.4 Conclusion

[Student 2 fills this in]

Chapter 3

[Student 3 Name] — [Student 3
Topic]

3.1 Methodology

[Student 3 fills this in]

3.2 Experiments & Findings

[Student 3 fills this in]

3.3 Discussion

[Student 3 fills this in]

3.4 Conclusion

[Student 3 fills this in]

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